Applied Sentience Labs Pvt. Ltd.

ASL-TR-2026-002 | Internal Research Document | Confidential

HiSE

Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

Research Foundation Document

Problem Statement · Abstract · Research Summary · Artifact Registry

Field	Detail
Document Type	Internal Research Foundation — Confidential
Prepared by	Nilesh Sanyasi, Founder & Lead AI Engineer, ASL
Institution	VPPCOE Mumbai — Final Year B.E. (AI & Data Science), 2026–27
Version	v1.0 — June 2026
Primary Target	ICNLP 2027 (IEEE Xplore) — Submission Deadline Nov 30, 2026
Research Code	ASL-TR-2026-002

Artifact Registry

Artifact	Name	Notes
Paper Title	HiSE: Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval	Primary submission: ICNLP 2027
Architecture / Mechanism	HiSA — Hierarchical Structural Addressing	The positional encoding mechanism; ASL's original contribution
Trained Embedding Model	HiSE-Embed	Open-source release on HuggingFace
Benchmark	StructEval-Bench v1	Open-source release on GitHub + HuggingFace
Retrieval System	StructRAG	Built on HiSE-Embed; production integration target for DataSense
GitHub / HuggingFace	asl-research/hise	Primary open-source repo
arxiv Preprint	Submitted parallel to conference	Increases visibility; does not conflict with IEEE submission

1. Problem Statement

1.1 The Core Observation

Every major embedding model in production today — BGE-M3, Qwen3-Embedding, E5-Mistral, Nomic Embed — encodes position the same way: as a linear integer. Token at position 1, token at position 2, token at position 47. The dominant mechanism doing this encoding is **Rotary Position Embedding (RoPE)**, which rotates query and key vectors by an angle proportional to their absolute sequence position, producing attention scores that reflect relative linear distance between tokens.

This works well for prose, conversation, and linear narrative. It works **poorly** for the dominant format of real-world enterprise and technical data: **hierarchically structured documents**. In these documents, meaning is not a function of token sequence position alone — it is a function of **structural address**: which section, which heading, which table row and column, which JSON key-path a token belongs to.

> **The Precise Failure**
> The sentence 'Revenue increased 40%' embedded under the heading '## Q3 Performance > Revenue' and the same sentence under '## Q3 Risks > Revenue' will produce near-identical embedding vectors in every current production model. The structural context that distinguishes them — one is a positive signal, one is a warning — is architecturally invisible. A RAG system

retrieving against a query 'what are the company's Q3 risks' may surface the performance-framed chunk as equally or more relevant than the risk-framed chunk. The LLM downstream then reasons over structurally wrong context and produces a confidently wrong answer.

1.2 Why This Has Not Been Solved

Three independent reasons, each sufficient on its own:

- **Research silo separation.** Table understanding researchers (TUTA, TAPAS, TabBiN) built structure-aware architectures for classification tasks — cell type classification, table type classification. Retrieval/embedding researchers (BGE, E5, Nomic) optimized for MTEB/BEIR benchmark scores. These two communities do not overlap, and MTEB/BEIR have no structurally-adversarial test cases that would expose the failure mode.
- **A misleading 2022 finding.** Wang et al. (2022) concluded 'Table Retrieval May Not Necessitate Table-specific Model Design,' suggesting generic text embeddings were competitive with structure-specific architectures on retrieval. This likely discouraged follow-up architecture work in the retrieval community — but the finding predates the RAG era and the more nuanced, adversarial structural failure modes that production RAG surfaces.
- **Benchmarks don't test structure.** If a benchmark has no structurally-adversarial pairs (same text, different structural context, opposite ground-truth relevance), no model will optimize for structural sensitivity, and no researcher will notice the gap. StructEval-Bench v1 is the first benchmark explicitly designed to test this — its existence is itself part of the research contribution.

1.3 The Gap, Stated Precisely

Research Gap

No embedding model trained for dense retrieval via contrastive learning incorporates a positional encoding mechanism that is sensitive to hierarchical structural address (document heading path, table row/column coordinates, JSON key-path depth) rather than linear sequence position alone. Every current production retrieval embedding model has this architectural blind spot, and no benchmark exists to measure it.

1.4 Structural Failure Modes — Evidence Table

Document Type	Structural Element	Concrete Failure	Consequence in RAG
Markdown / Technical Docs	Heading hierarchy (H1 > H2 > H3)	'pip install x' under ## Installation and ## Troubleshooting embed identically	Model retrieves installation instructions for a troubleshooting query
Financial Reports	Section hierarchy (Performance vs Risks)	'Revenue grew 40%' under ## Performance and ## Risks embed identically	Risk analysis query surfaces positive-framed content; analyst gets wrong signal

Document Type	Structural Element	Concrete Failure	Consequence in RAG
Tables / Spreadsheets	Row × Column coordinate	Cell value '47' has no row/column anchor; headers and values detached in vector space	Query about Q3 revenue surfaces Q3 loss row; numerical analysis breaks
JSON / Dict / Config	Key-path nesting (parent.child.key)	config.primary.timeout and config.fallback.timeout embed near-identically	System config query returns fallback values as primary; silent misconfiguration
Legal / Contracts	Clause hierarchy (Article > Exception)	Exception clause text embeds close to obligation clause text	Legal query retrieves exception when obligation is needed; compliance risk

2. Abstract

HiSE: Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

Nilesh Sanyasi — Applied Sentience Labs Pvt. Ltd., Mumbai, India

Dense retrieval systems depend on embedding models to produce vector representations that preserve semantic meaning across queries and documents. Current embedding models encode token position using linear mechanisms — predominantly Rotary Position Embedding (RoPE) — which capture sequential distance between tokens but are architecturally blind to **hierarchical structural position**: the heading path, table row-column coordinate, or JSON key-path that governs a token's contextual meaning in structured documents. This produces a systematic failure mode in which tokens with near-identical surface text but structurally-opposed meanings — a metric framed as a performance indicator versus a risk indicator, a configuration value under a primary versus fallback key — are assigned dangerously similar embedding vectors, causing retrieval systems to surface structurally incorrect context for downstream language model reasoning.

We present **HiSE** (Hierarchical Structure-aware Embeddings), a contrastive retrieval embedding model incorporating **HiSA (Hierarchical Structural Addressing)**: a novel positional encoding mechanism that extends RoPE from one-dimensional sequence position to multi-dimensional structural address, encoding depth, parent path, and sibling index as explicit rotational components of the position representation. HiSE is trained from a small SLM backbone via contrastive learning with structurally-adversarial hard negatives — pairs that are surface-text-similar but structurally-opposed — forcing the model to develop structural sensitivity as a first-class representational property rather than an emergent side effect.

We validate HiSE on **StructEval-Bench v1**, the first retrieval benchmark explicitly designed to test structural discrimination: 300–500 structurally-adversarial pairs across three document formats (markdown heading hierarchies, tabular row-column structures, and nested JSON key-paths), with ground-truth relevance labels requiring structural sensitivity to resolve. We additionally evaluate on standard BEIR and MTEB retrieval subsets to confirm that structural gains do not degrade general retrieval performance. HiSE is compared against same-size standard-architecture baselines trained on identical data, isolating the HiSA positional encoding as the experimental variable. Ablations confirm that the architectural modification — not data distribution — drives structural retrieval improvement.

HiSE-Embed and StructEval-Bench v1 are released open-source. HiSA is designed as a format-agnostic structural addressing mechanism applicable to any document format representable as a tree, providing a principled positional foundation for structure-sensitive retrieval beyond the three validated domains.

3. Research Summary

3.1 What We Are Building

Component	What It Is	What Makes It Novel
HiSA (Hierarchical Structural Addressing)	A positional encoding mechanism that assigns every token a structural address — depth in document tree, parent section identity, sibling index — and encodes this as multi-dimensional rotational components, extending RoPE from 1D to structural N-D	No prior work applies tree-positional encoding to contrastive retrieval embedding models. TUTA/DOM-LM applied structural encoding to classification tasks only. RethinkRoPE (2025) extended RoPE to N-D for vision/spatial — not applied to document hierarchy or retrieval.
HiSE-Embed (The Trained Model)	A small (~100-300M parameter) SLM-backbone encoder with HiSA replacing standard RoPE, trained via contrastive learning with structurally-adversarial hard negatives	First contrastive retrieval model with explicit hierarchical positional encoding validated on structurally-adversarial retrieval benchmarks
StructEval-Bench v1 (The Benchmark)	300-500 structurally-adversarial pairs across markdown, tables, and JSON — same or similar surface text, structurally-opposed ground-truth relevance. Metrics: SDS (Structural Discrimination Score) and RIR (Retrieval Inversion Rate)	First retrieval benchmark designed specifically to measure structural sensitivity. BEIR/MTEB have no structurally-adversarial test cases.
StructRAG (The Retrieval System)	A retrieval pipeline built on HiSE-Embed with structure-preserving chunking that maintains heading/table/key-path metadata through to encoding	Demonstrates end-to-end production viability; serves as integration point for ASL DataSense

3.2 The Architectural Contribution — HiSA in Plain Terms

Standard RoPE assigns every token one number — its sequence position — and rotates its query/key vectors by an angle proportional to that number. The result: attention scores reflect how far apart tokens are in the token stream. Two tokens 10 positions apart get the same attention relationship regardless of whether they are in the same section or different sections.

HiSA replaces that one number with a **structural address tuple**: (sequence_position, depth, parent_id, sibling_index). Each dimension gets its own rotational component, applied jointly so that the final attention score reflects both sequential proximity and structural proximity. Two tokens in the same heading section get a structurally-close representation even if they are 200 tokens apart sequentially. Two tokens in sibling sections under different parents get a structurally-distant representation even if they are adjacent in the token stream.

Standard RoPE (current models)	HiSA (our contribution)
Attention(token_A, token_B) ~ f(|pos_A - pos_B|) Only one signal: how far apart in the token stream.	Attention(token_A, token_B) ~ f(|pos_A - pos_B|, depth_dist, parent_match, sibling_dist) Four signals: sequence + structural address.

3.3 Prior Art Map — Where We Fit

Work	What It Does	Why It's Not Enough	Our Relationship
TUTA (Microsoft, KDD 2021)	Bi-dimensional coordinate tree for table structure; tree-based attention and position embedding	Trained for cell/table type classification only. No contrastive retrieval training. No cross-format claim.	Foundational motivation; we adapt the structural coordinate concept to RoPE for retrieval
DOM-LM (2022)	Injects node index, parent index, depth, sibling index as additive position embeddings into BERT for HTML	HTML-only. Additive not rotational. Not trained for retrieval/RAG use case.	Most direct architectural prior art; we extend the idea to RoPE and to retrieval training
Tree-Enhanced CodeBERTa (2025)	Integrates AST depth and sibling index as positional embeddings into CodeBERTa for code representation	Code/AST only. Classification tasks. Not a contrastive retrieval model.	Validates that tree PE improves transformer representations; we apply to document retrieval
Struc-EMB (Georgia Tech, Oct 2025)	Tests sequential concatenation vs parallel caching of structural context in LLM-based embedders	Tests prompt/context-level approaches, not architectural position encoding changes. No structural hard negatives in training.	Directly motivates why architecture-level fix beats prompt-level workaround
RethinkRoPE (arXiv 2504.06308, 2025)	Mathematical blueprint for extending RoPE to N-dimensional positional embedding	Proposed for vision/spatial domains. Not applied to document hierarchy or retrieval.	Mathematical foundation for HiSA's multi-dimensional rotational extension
TabBiN (EDBT 2025)	Bi-dimensional tabular coordinates and visibility matrix for 2D tabular data embedding	Table-specific. Not designed for dense retrieval RAG. No cross-format generalization.	Validates that bi-dimensional structural encoding improves table representations
BGE-M3, Qwen3-Embedding, Nomic Embed	Current SOTA retrieval embedding models	Linear RoPE only. No structural address. Fail on structurally-adversarial pairs (confirmed by failure-mode survey).	Primary baselines; same-size variants used in ablation study

3.4 What We Prove — The Experimental Claim

Primary Hypothesis (Falsifiable)

A contrastive retrieval embedding model incorporating HiSA (hierarchical structural addressing via multi-dimensional RoPE extension) will achieve statistically significant improvement in structural discrimination (SDS) and retrieval inversion rate (RIR) on StructEval-Bench v1 compared to a same-size same-data same-compute baseline using standard linear RoPE — and will not degrade general retrieval performance (nDCG@10) on BEIR/MTEB subsets beyond margin of error.

The ablation design: same SLM backbone, same training data, same compute budget, same contrastive loss — only the positional encoding mechanism changes. This isolates HiSA as the experimental variable and makes the result unambiguous.

3.5 Why This Is Not Integration Engineering

The three document formats (markdown, tables, JSON) are not three separate contributions assembled into one paper. They are three experimental domains that share a single structural property — all three are trees — used to validate that HiSA's positional encoding is **format-agnostic**, not format-specific. The contribution is the mechanism (HiSA) and the claim (linear position is the wrong inductive bias for hierarchical documents). The formats are evidence for the generality of that claim.

Integration Engineering (what we are NOT doing)	**Representational Research (what we ARE doing)**
Combine markdown encoder + table encoder + JSON encoder into one system. Contribution = unified pipeline. Reviewer reads: 'Systems engineering.'	Propose one positional encoding mechanism (HiSA) that handles all tree-structured documents. Validate on three domains to prove format-agnosticism. Reviewer reads: 'Foundational positional theory with strong empirical validation.'

4. Research Context & Motivation

4.1 Origin — DataSense & Production RAG Failure

This research originated from a production failure observed during development of **DataSense**, ASL's AI-powered data analysis product for SMBs. DataSense's current architecture (v1.0, March 2026) compensates for AI models' inability to reason over structured data by generating an explicit 'data manifest' — a statistical description of column types, relationships, and domain signals — passed to the model as prompt context.

This is a prompt-engineering workaround for a representational problem. The correct fix is not a better prompt — it is an embedding model that encodes structural context natively, so that retrieval over structured documents (financial tables, configuration files, technical specifications) surfaces the right structural chunk without requiring an explicit manifest intermediary. HiSE-Embed is that fix. StructRAG is its integration point into DataSense Phase 2.

4.2 Placement in the Research Landscape

The embedding model field's 2023–2025 focus was scale and context length (BGE-M3, E5-Mistral, Qwen3-Embedding) competing on MTEB leaderboard scores. A 2025 Google DeepMind paper explicitly flagged theoretical limitations of single-vector embedding-based retrieval, noting that models increasingly face fundamental representational ceilings as the diversity of content to represent grows. HiSE addresses one specific, well-defined ceiling: structural sensitivity. It is not a general-purpose improvement — it is a targeted fix for a measurable, demonstrated failure mode.

4.3 The LLM Extension (Future Work)

If HiSA improves retrieval embeddings by making the model structurally sensitive to its input, the same mechanism applied to a **generative LLM's** positional encoding would make the model structurally sensitive to its context window during generation — attending differently to tokens in the same structural section versus structurally-opposed sections, even at the same sequential distance. This is a legitimate and interesting extension (active work on RoPE modification for LLMs exists in 2024–2025) but requires a full generative LLM training run and is explicitly scoped as future work in the paper. It is named here to establish the research trajectory.

5. Publication Strategy

Priority	Venue	Deadline	Strategy
Primary	ICNLP 2027 (IEEE Xplore)	Nov 30, 2026	Main submission. Best timeline match. NLP + IR scope fits perfectly.

Priority	Venue	Deadline	Strategy
Backup 1	PUNECON 2026 (IEEE Pune Section)	~Sep–Oct 2026	India-based. Strong for college final review. Check punecon.ieeepunesection.org for exact deadline.
Backup 2	MLNLP 2026 (IEEE)	Check IEEE CIS calendar	ML + NLP focus. Confirms topic fit if ICNLP rejects.
Parallel	arXiv preprint	Anytime post-submission	Post simultaneously with conference submission for visibility. Does not conflict with IEEE.
Stretch	ACL/EMNLP Workshop	Mar–May 2027	If results are strong enough post-primary submission. Workshop track receptive to novel systems work.
Second Paper	StructEval-Bench standalone	2027	If benchmark results are rich enough to stand alone as a dataset/evaluation contribution paper.

Rule: sequential submissions only, never simultaneous. One venue at a time. Rejected paper gets reviewer feedback incorporated before next submission.

6. Open Source & ASL Research Identity

All research artifacts release open-source under the **asl-research** GitHub organization. This builds ASL's public research identity — the same strategy that allowed small labs like EleutherAI and MosaicML to attract research talent and investment before being acquired.

Artifact	Release Platform	License	Timing
HiSE-Embed (trained model)	HuggingFace: asl-research/hise-embed	Apache 2.0	With paper submission (or arXiv preprint)
StructEval-Bench v1 (benchmark)	HuggingFace Datasets + GitHub	CC BY 4.0	With paper submission
HiSA implementation (code)	GitHub: asl-research/hise	MIT	With paper submission
StructRAG pipeline	GitHub: asl-research/structrag	Apache 2.0	Phase 5 completion
Training code & configs	GitHub: asl-research/hise	MIT	With paper submission